

Project Instruction

“Now is the moment to apply what you’ve learned”

From one **decision tree** to a **random forest**,
from a single model to an **ensemble of solutions**.

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Welcome to the ML Project Challenge!

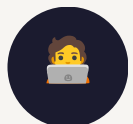
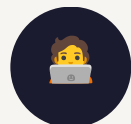
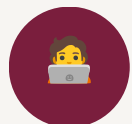
The project is designed to give you hands-on experience with real-world machine learning problems.

What You'll Experience

- Working with messy, real-world datasets
- Building ML models across 5 different tasks
- Making decisions about model selection & evaluation
- Collaborating in a team like industry professionals

You'll Work in Groups of 5!

- Real ML projects require teamwork
- Divide work efficiently across 5 tasks
- Learn from each other's approaches
- Practice collaboration skills



Real ML projects require teamwork — divide, learn, and build together.

Competition Enrollment

1 Team Register

- Fill in team info on the Google Sheet.
- Your “Team ID” will be the official Team name.

2 Join Competition

- Create or log in to your Kaggle account
- Register for the competition using the Kaggle link.

3 Team Setup

- Name your Kaggle team with the exact “Team ID” (e.g., ML001).
- Invite all teammates to the Kaggle team.

4 Participation

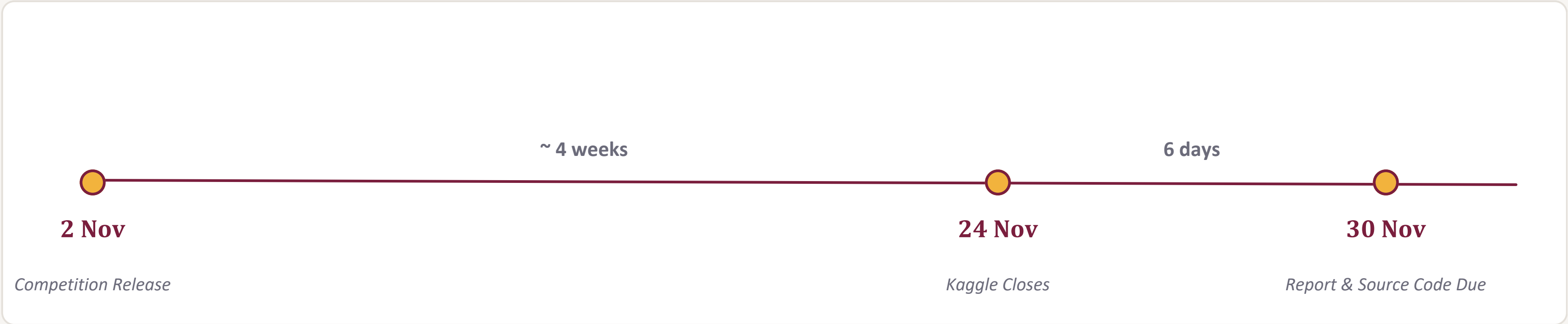
- Download the dataset
- Train your best model
- Upload your submission
- Check the leaderboard.

Grading Criteria & Timeline

Performance on Kaggle leaderboard	70%
Technical Report	20%
Source Code	10%

“The technical report and source code submission will be posted on LEB2”

Timeline



Submission Checklist

1. Kaggle

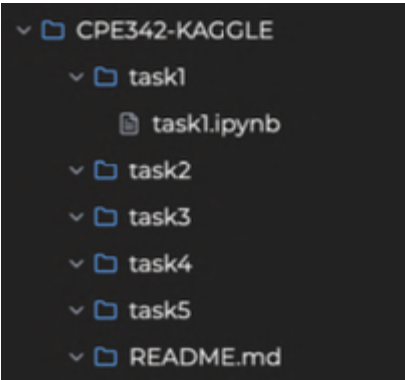
- Join the competition
- Register with Kaggle account
- Fill in team info on Google Sheet
- Name Kaggle team with exact Team ID
- Invite all teammates
- Download dataset & train model
- Submit predictions to Kaggle

2. Technical Report

- Methodology per task (e.g., EDA, preprocessing, model design, results)
- Insights & common mistakes
- Domain interpretation
(link to business impact)
- Team collaboration – roles & takeaways
- Conclusion – lessons learned, ML insights, future improvements
- Language: English or Thai

3. Source Code

- Create a GitHub repository
- Organize code into folders per task (Task1, Task2, ...)
- Example structure:



How Your Kaggle Score Converts to a Grade

Above Baseline (guarantee grade at $\alpha\%$)

$$\text{Grade}\% = \alpha + [(100\% - \alpha) \times \Phi(t)]$$

T-Score $t = (\text{Your Score} - \mu) / \sigma$
 μ = mean of scores above baseline σ = std. dev. of scores above baseline
 $\Phi(t)$ = Standard Normal CDF (Z-table)

Baseline — designed to be achievable by everyone

Below Baseline

$$\text{Grade}\% = \alpha \times (\text{Your Score} / \text{Baseline})$$

Example grading shown uses $\alpha = 60\%$.

Example Grading ($\alpha = 60\%$)

Rank	Score	Grade%
1	0.650	98.65
2	0.620	97.51
5	0.520	88.50
8	0.460	79.61
11	0.410	72.17
14	0.350	65.45
Baseline	0.300	—
17	0.280	56.00
19	0.200	40.00
20	0.150	30.00

Need a Hand? We're Here to Help!

Your Teaching Assistants (TAs) are your first stop for questions, road bumps, and feedback. We're excited to support your team!

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TIME TO REVEAL THE PROJECT...

Karena Player Intelligence System

ML meets Game Industry

This isn't just about algorithms — it's about solving business problems with data.

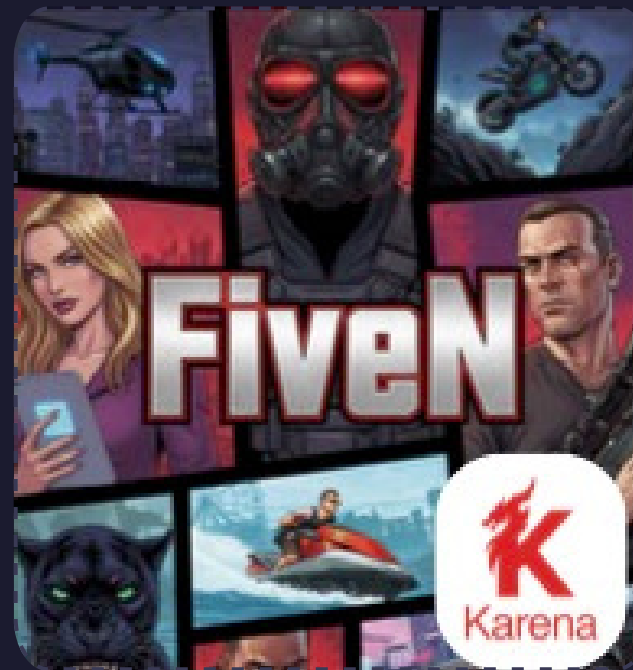


Karena

Situation

Your team consists of Machine Learning Engineers at **Karena Thailand** — a gaming platform company (similar to Garena) that develops and publishes multiple online games, for example PUBE Mobile, Free Fried, FiveN, Roblock, and 7 Knights.

Publishing Games



The Challenge

Karena needs an AI-powered Player Intelligence System to:



Maintain

game integrity across all
platforms



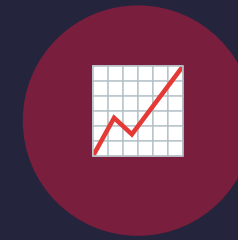
Understand

player behavior and
preferences



Optimize

monetization strategies

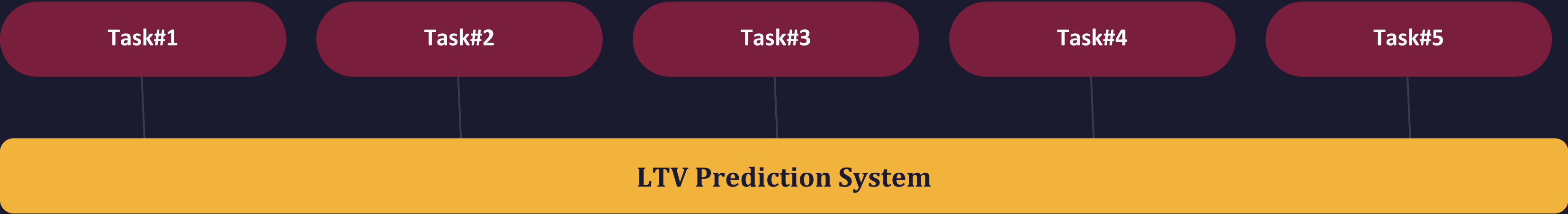


Improve

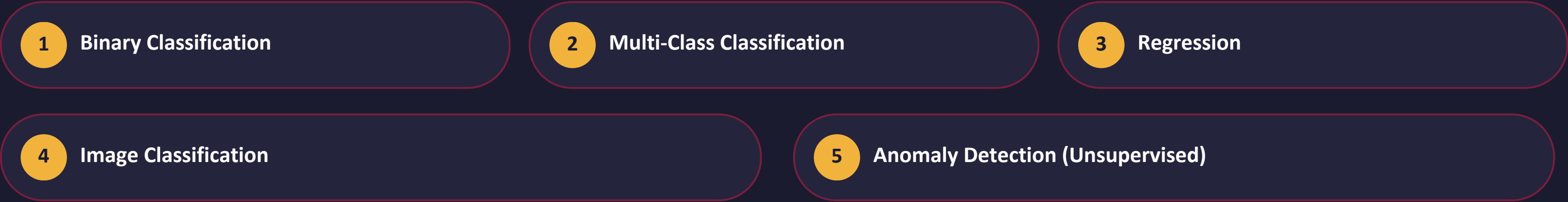
player retention and lifetime
value

Your Team's Mission

Build a comprehensive ML pipeline with 5 tasks that feed into a final Player Lifetime Value (LTV) Prediction System to help the business make data-driven decisions.



What You'll Build — 5 Interconnected Models

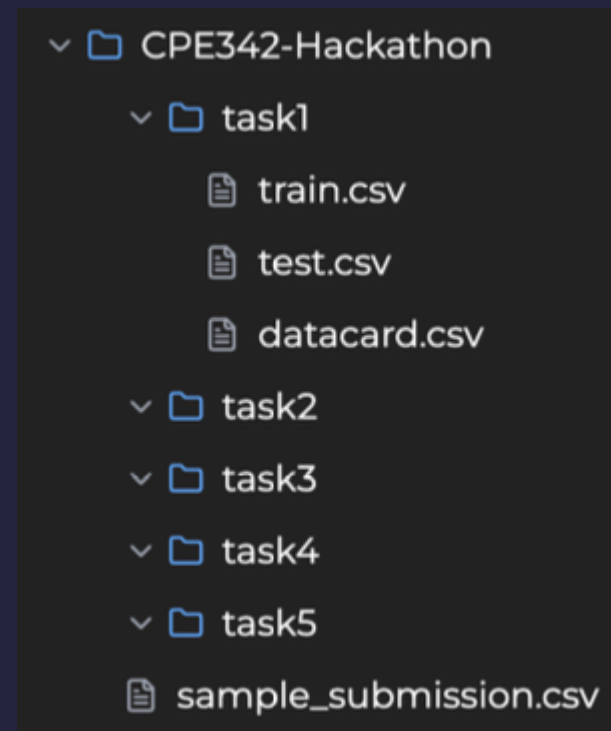


Instruction Structure & Competition Rules

In Each Task Instruction

- Situation
- Mission
- Task Category
- Evaluation Metric
- Example Features
- Hints & Best Practices

Dataset structure per task
(Kaggle download):



Competition Rules

Data

Use provided datasets only. Combine Train+Val for training.
No Test-set usage for training (no pseudo-labeling / fine-tuning).

Model Development

Task 4 allows pre-trained models; Tasks 1,2,3,5 train from scratch. Feature engineering & ensembling encouraged.
AutoML frameworks prohibited; hyperparameter tuning tools allowed.

Collaboration

Work only with your team. Concept discussion OK, but no sharing code or predictions.
All members must contribute meaningfully.

TASK 1

Task 1: Anti-Cheat Pre-Filter

Situation

Karena Thailand receives over 10,000 cheat reports daily across all games, but the manual review team can only handle 500 cases per day, creating a massive backlog of 45,000+ pending reports. Many reports are false positives from frustrated players who lost matches fairly, while real cheaters continue to damage game experience and drive legitimate players away. The current automated system has a 70% false positive rate, meaning innocent players get flagged and sometimes banned, leading to customer service nightmares and permanent player loss. Meanwhile, sophisticated cheaters who use subtle aimbots [1] or wallhacks [2] slip through undetected. Player retention has dropped 15% in competitive game modes due to cheating complaints, directly impacting revenue by approximately \$15 million daily. The company needs an intelligent pre-filtering system that can accurately identify high-probability cheaters for priority manual review, reducing workload while catching real threats

[1] Aimbot: Cheating software that automatically aims at enemies

[2] Wallhack: Exploit allowing players to see through walls

Binary Classification



TASK 1

Anti-Cheat Pre-Filter

Your Task

Build a binary classifier to automatically pre-filter reports and prioritize high-confidence suspicious accounts for manual review.

ML Task

Binary Classification

Metric: F2

$$F2 = 5 \cdot (P \cdot R) / (4P + R)$$

Target: `is_cheater` (0=Legitimate, 1=Cheater)

Features Example

- `kill_death_ratio` (float): kills-to-deaths ratio
- `headshot_percentage` (float): % kills that are headshots
- `win_rate` (float): overall win rate %
- `accuracy_score` (float): shot accuracy %
- `reaction_time_ms` (float): avg. reaction time (ms)
- 33 features total (32 predictive + target)



Hints & Best Practices

- Handle missing values — impute or drop carefully
- Handle class imbalance: resampling, class weights, SMOTE
- Feature engineering — build interaction features
- Use cross-validation for reliable performance
- Try decision-threshold tuning
- Ensemble models for robustness

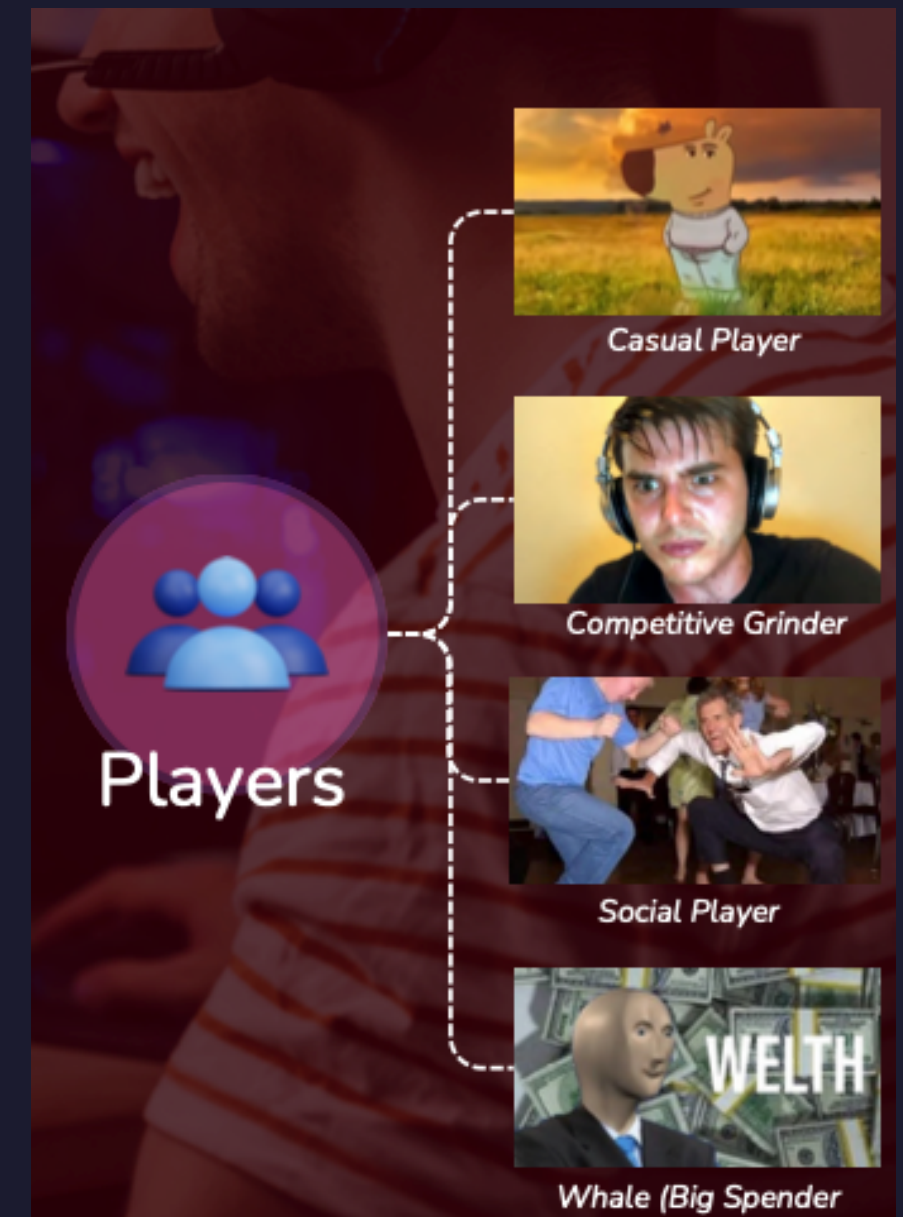
TASK 2

Task 2: Player Segment Classification

Situation

Karena Thailand's marketing team currently runs generic campaigns across all players, resulting in extremely low engagement rates of only 2-5%. They spend approximately \$60 million monthly on promotional campaigns, but most fall flat because they're not personalized. For example, competitive esports tournament announcements are shown to casual mobile players who just want to relax, cosmetic item sales are advertised to players who have never made a purchase, and FiveN roleplay event invitations go to hardcore PUBE competitive grinders. This misalignment wastes massive advertising budgets and annoys players with irrelevant content. The top 5% of players (whales) generate 60% of total revenue, but the company can't identify them reliably using simple metrics like playtime alone. A high-playtime player could be a competitive grinder seeking glory, a social player enjoying friend time, or a whale collecting items—each needs completely different messaging. Proper segmentation could increase campaign effectiveness by 3-4x, translating to millions of baht in additional revenue and improved player satisfaction.

Multi-Class Classification



TASK 2

Player Segment Classification

Your Task

Classify each player into one of four behavioral segments to enable targeted marketing and personalized experiences.

ML Task

Multi-class Classification

Metric: F1

$$F1 = 2 \cdot (P \cdot R) / (P + R)$$

Target: segment (0=Casual, 1=Grinder, 2=Social, 3=Whale)

Features Example

- play_frequency (float): avg. days/week played
- avg_session_duration (float): avg. session length (min)
- total_spending_thb (float): lifetime spending
- avg_monthly_spending (float): average monthly spend
- friend_count (float): number of in-game friends
- 45 features total (44 predictive + target)



Hints & Best Practices

- Leverage Task 1 tips — missing values & imbalance
- Use Stratified K-Fold to preserve class distribution
- Try XGBoost, LightGBM, or CatBoost
- Stack these models together for extra robustness
- Engineer interaction & ratio features

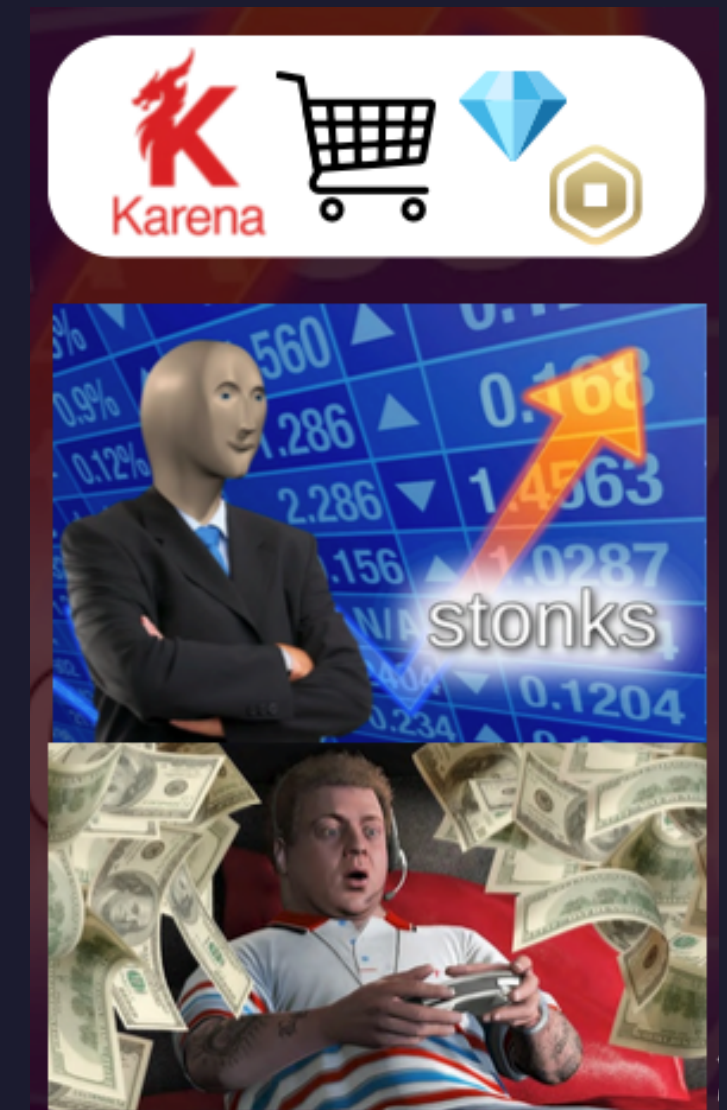
TASK 3

Task 3: Player Monthly Spending Prediction

Situation

Karena Thailand's finance team struggles with revenue forecasting, consistently missing quarterly targets by 35-40%, which impacts budgeting, hiring decisions, and investor confidence. Without accurate player-level spending predictions, multiple departments are operating blindly. The VIPsupport team of 20 people provides white-glove service but doesn't know which players actually deserve priority treatment—last quarter they spent significant resources on 150 high-playtime players who collectively spent only ฿12,000, while a whale who spent ฿255,000 waited 3 days for ticket response and subsequently left for a competitor. The marketing team sends promotional discount codes randomly, often giving 50% off to whales who would have paid full price anyway, resulting in ฿6 million+ monthly revenue loss. Customer service wastes time on players unlikely to ever monetize while high-value customers receive generic treatment. Additionally, the company needs to optimize their limited-time event scheduling and new content releases based on when high-spending players are most active. Accurate spending predictions would enable proper resource allocation, targeted offers, revenue forecasting, and VIP program optimization across the organization.

Regression



TASK 3

Player Monthly Spending Prediction

Your Task

Predict how much (in THB) each player will spend on in-game currency and items in the next 30 days.

ML Task

Regression

Metric: Normalized MAE

$$\text{NMAE} = 1 / (1 + \Sigma |y - \hat{y}| / \Sigma y)$$

Target: `spending_30d` (continuous, THB)

Features Example

- `friend_count` (float): number of in-game friends
- `event_participation_rate` (float): % limited-time events joined
- `daily_login_streak` (float): current login streak
- `avg_session_length` (float): avg. session duration (min)
- `historical_spending` (float): past spending
- 33 features total (32 predictive + target)



Hints & Best Practices

- Zero-inflated data — try a two-stage model:
(1) will spend? (2) how much?
- Feature engineering & log transformation of target
- Try XGBoost / LightGBM / CatBoost Regressors
- Ensemble / stacking for extra accuracy
- Watch for common mistakes in target scaling

TASK 4

Task 4: Game Title Detection

Situation

Karena Thailand operates five distinct games but can't automatically identify which game is being played from screenshots. The customer support team receives 2,000 daily tickets with images and wastes 10+ minutes per ticket figuring out which game before routing to specialists. The "Play Multiple Games for Rewards" campaign can't verify 50,000+ screenshot submissions, forcing manual checks of only 3% (leading to fraud). Players post 100,000+ gameplay images weekly across platforms that can't be auto-tagged for contests or moderation. Marketing can't track which games are trending without manual review. An automated detection system would enable faster support, verified campaigns, scalable content moderation, and cross-game insights.

Image Classification



TASK 4

Game Title Detection

Your Task

Build an image classifier that identifies which Karena game is being played from a single screenshot.

ML Task

Image Classification (Multi-Class)

Metric: F1 (Macro)

$$F1 = 2 \cdot (P \cdot R) / (P + R)$$

Target: `game_title` (5 classes)

Features Example

- Gameplay screenshots (144p, centered crop to 230x120)
- Sources: live streams, submissions, forums, tickets
- Variety: different maps, phases, weather, UI states
- 5 classes: 7-11K, fiveN, freefried, pube, roblock
- Pre-trained models allowed for this task



Hints & Best Practices

- Review image classification basics & data augmentation
- Watch for class imbalance in images
- Try VGG16/VGG19, Vision Transformer (ViT), Swin Transformer
- Use cross-validation for CNNs; ensemble methods help
- You may need a GPU — Colab/Kaggle offer free T4 GPUs
- Kaggle: ~30 GPU hrs/week; Colab: dynamic, no fixed limit

TASK 5

Task 5: Account Security Monitoring

Situation

Karena Thailand faces over 5,000 confirmed account thefts monthly (actual number likely 2x higher), making it the #1 customer support issue consuming 60% of team resources. Beyond theft, sophisticated account selling rings destroy competitive integrity, bot farms operate 24/7 causing game economy inflation, and players exploit VPNs for regional pricing (70% discounts), costing millions of baht annually. The current rule-based security system has 60% false positive rate—flagging legitimate travelers while missing smart attackers who wait 24 hours between actions. Simple thresholds flag players who legitimately improve while missing gradual botting patterns. The security team drowns in alerts while real threats slip through. The company needs an intelligent unsupervised anomaly detection system that learns normal behavior patterns and flags truly unusual activities without rigid rules or labeled data.

Anomaly Detection



TASK 5

Account Security Monitoring

Your Task

Build an unsupervised anomaly detection model to flag accounts with unusual behavior, using techniques like Isolation Forest or Autoencoders — without relying on labeled training data.

ML Task

Anomaly Detection (Unsupervised)

Metric: F3

$$F3 = 10 \cdot (P \cdot R) / (9P + R)$$

Target: `is_anomaly` (0=Normal, 1=Anomalous)

Features Example

- `login_count_1/2/3/4` (float): logins per time period
- `login_lat/lon_1/2/3/4` (float): login coordinates
- `location_changes_1/2/3/4` (float): significant location changes
- `device_count_1/2/3/4` (float): unique devices used
- 33 features total (32 predictive + target)
- Note: no labeled training set provided — test.csv only



Hints & Best Practices

- Try Isolation Forest, One-Class SVM, or Auto-Encoder
- Even unsupervised models can be ensemble
- Read up on soft vs. hard voting techniques
- Tune contamination / threshold carefully

Submission Format

Fill your answer in sample_submission.csv

id	task1	task2	task3	task4	task5
ANS00001	0	2	0	freefried	1
ANS00002	1	3	10000	fiveN	0
ANS00003	...	...	...	...	...
ANS25889	...	...	...	...	...

One prediction column per task — fill every row for every task.

25,889 rows × 6 columns

Evaluation Metrics

F1

$$F1 = 2 \cdot (P \cdot R) / (P + R)$$

Applied: Task 2, Task 4

F2

$$F2 = 5 \cdot (P \cdot R) / (4P + R)$$

Applied: Task 1

F3

$$F3 = 10 \cdot (P \cdot R) / (9P + R)$$

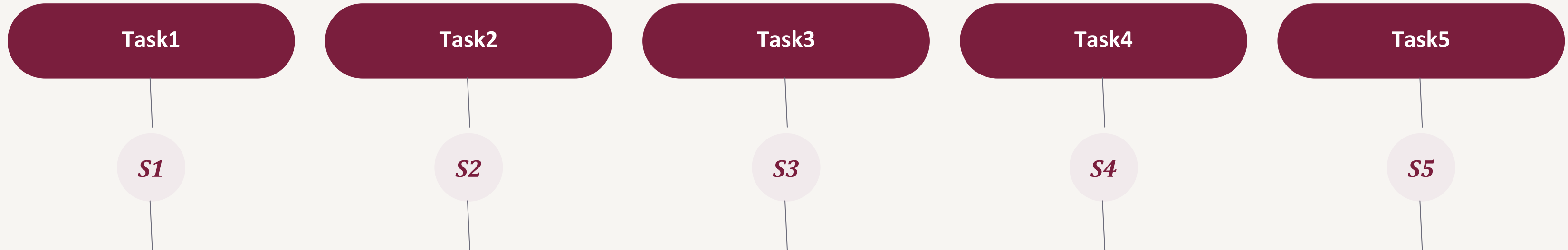
Applied: Task 5

NMAE

$$1 / (1 + \Sigma |y - \hat{y}| / \Sigma y)$$

Applied: Task 3

Kaggle Score Calculation



$$\text{Weighted Score} = (w_1S_1 + w_2S_2 + w_3S_3 + w_4S_4 + w_5S_5) / \sum w_i$$

range: [0, 1]

 **Leaderboard**

Each task is evaluated independently by its own metric, then combined into one weighted score that ranks your team on the Kaggle leaderboard.

BASELINE

Baseline Model

We built the first baseline using only the models we covered in class.

Weighted Score (Public) = 0.68

Ground Baseline

Logistic Regression

Random Forest

CNN

Isolation Forest

The challenge baseline selects a model and leverages techniques from the hints provided in this instruction slide.

Weighted Score (Public) = 0.75

Challenge Baseline

To go beyond the baseline, try experimenting with cutting-edge models and techniques. We encourage you and your team to explore and experiment with new ideas!

let's get our hands dirty!

“Good luck!” – Aj & TA